

LEMON-Factor: Bidirectional Factor-Level Semantic Alignment for Graph–Text Pairs

Tomilov A.A.¹, Gineva D.², and Tirskih D.²

¹ Independent researcher, affiliation to be confirmed

² ITMO University, St. Petersburg, Russia

Abstract. Graph–text evaluation requires more than checking whether the right entity names occur in a fluent sentence. A faithful verbalization must preserve the roles and relation meanings encoded by graph predicates, and a useful text-to-graph reconstruction must recover the same predicate-level information. We propose LEMON-FACTOR, a bidirectional factor-level diagnostic framework for graph–text pairs. LEMON-FACTOR decomposes RDF predicates into weighted semantic factors and evaluates both $\text{KG} \rightarrow \text{Text}$ coverage and $\text{Text} \rightarrow \text{KG}$ recoverability. The framework is positioned between triple-level factuality metrics, which typically issue one decision per triple, and atomic-fact metrics, which decompose generated text rather than source predicates. On a stratified WebNLG pilot, expanded Forward LEMON-FACTOR reaches 0.7915 and Reverse LEMON-FACTOR reaches 0.7789. A MINE-STYLE adaptation inspired by KGGen’s Measure of Information in Nodes and Edges shows that node information is easier to recover than edge information: on a locked 50-fact subset, deterministic node information is 0.9600, edge information is 0.5400, and LLM-judged fact recoverability is 0.5200. Controlled perturbation calibration further shows that relation deletion reduces Forward LEMON-FACTOR by 0.211 while exact entity-label coverage changes by only 0.002. These results support LEMON-FACTOR as an interpretable pilot framework for diagnosing relation-level semantic alignment. We treat rule-based expansion and synthetic LLM adjudication as provisional resources, not as human expert validation.

Keywords: Graph-to-text evaluation · Semantic alignment · Knowledge graphs · WebNLG · Predicate decomposition · Evaluation calibration

1 Introduction

Knowledge graphs encode information as typed relations between entities. In RDF-style graph-to-text generation, an input edge (s, p, o) is not only a pair of entity names. The predicate p states which relation holds, which semantic roles the arguments fill, and which part of the meaning must be preserved in text. The same issue appears in the reverse direction. If a text is converted back into a graph, a useful reconstructed edge must preserve relation meaning, not merely mention the same entities.

This distinction is often hidden by holistic automatic metrics. A sentence may mention *Alan Bean* and *Wheeler, Texas* while omitting that the relation is *birthPlace*. It may paraphrase the relation, replace it with a weaker predicate, or preserve only the subject and object. Standard WebNLG evaluation uses reference metrics such as BLEU, METEOR, chrF++, TER, BERTScore, and BLEURT, together with human criteria such as data coverage, relevance, correctness, and fluency [1–3]. These criteria are useful, but they do not say which internal semantic component of a predicate is covered or missing.

We propose LEMON-FACTOR, a source-driven diagnostic framework for graph–text semantic alignment. The core idea is to decompose each source predicate into typed, weighted semantic factors. For example, *birthPlace* can be represented by a person-like subject factor, a biographical-relation factor, and a place-like object factor. LEMON-FACTOR then checks whether these factors are expressed in text or recovered in a reconstructed graph. This creates a diagnostic layer between binary triple-level metrics, such as FactSpotter, and text-driven atomic-fact metrics, such as FactScore [10, 11].

The framework is bidirectional. In the forward direction, $\text{KG} \rightarrow \text{Text}$, it measures whether a text covers the semantic factors of the input graph. In the reverse direction, $\text{Text} \rightarrow \text{KG}$, it measures whether a reconstructed graph preserves the semantic factors of the source graph. We also implement a MINE-STYLE node/edge baseline inspired by KGGen’s Measure of Information in Nodes and Edges (MINE), which evaluates text-to-KG information retention through recovered nodes and relations [16]. The baseline is used as a transparent comparison point; it is not a reproduction of the full KGGen/MINE benchmark.

The paper asks four research questions. **RQ1:** Does factor-level coverage capture graph-to-text semantic preservation beyond entity-label overlap? **RQ2:** Does the same factor representation support both forward coverage and reverse recoverability? **RQ3:** How does factor-level recoverability relate to node/edge information retention in a MINE-style baseline? **RQ4:** Do the metrics behave predictably under meaning-destroying and meaning-preserving perturbations?

The contributions are: (1) a bidirectional, weighted factor-level formulation for graph–text semantic alignment; (2) a reproducible WebNLG pilot with factor decomposition resources, forward and reverse scores, and audit trails; (3) a MINE-style node/edge information baseline adapted to WebNLG graph–text pairs; and (4) controlled perturbation calibration showing that relation-level damage is detected by factor-aware metrics while simple entity-label overlap remains nearly unchanged. The study is deliberately framed as a pilot. Factor decompositions are bootstrapped with rules and synthetic LLM adjudication, and the resulting resource is not claimed to be human-validated gold.

2 Related Work

2.1 RDF-to-text benchmarks and reference metrics

The WebNLG Challenge established RDF-to-text generation as a benchmark task in which systems map DBpedia-derived RDF triples to natural language

[1]. The WebNLG+ shared task extended the setting to bilingual and bidirectional generation and included both automatic and human evaluation [2]. The GEM benchmark further argued for broader evaluation practice and challenge sets for NLG evaluation [3]. The standard WebNLG automatic stack includes reference-based metrics such as BLEU, METEOR, chrF++, TER, BERTScore, and BLEURT [4–6]. These metrics produce useful holistic similarity signals, but they do not localize errors to a source predicate or to a semantic role inside the predicate.

2.2 Source-aware coverage and factuality metrics

A second line of work evaluates generated text against input data. PARENT weights n-gram precision and recall by evidence in the source table and reference text [7]. Data-QuestEval adapts question-answering evaluation to data-to-text semantic coverage without requiring a reference [8]. AlignScore learns a unified alignment function for factual consistency across several supervision sources [9]. The closest graph-to-text predecessor is FactSpotter, which decides whether a given triple (s, p, o) is present in a generated text and reports strong correlations with human judgments for data correctness, coverage, and relevance [10]. LEMON-FACTOR complements this line by moving from a binary whole-triple decision to a weighted decomposition of the predicate into semantic factors.

2.3 Atomic facts, semantic graphs, and directionality

Atomic factuality metrics decompose generated text into verifiable claims. FactScore evaluates long-form text by decomposing it into atomic facts and checking support against a knowledge source [11]; VERIScore extends claim verification for long-form factuality [12]. This direction is text-driven: the output text is decomposed into claims. LEMON-FACTOR is source-driven: the source predicate is decomposed into factors that can be checked in text or in a reconstructed graph.

Semantic graph metrics such as Smatch compare graph structures, especially AMR graphs, by maximizing overlap under node alignment [13]. S²match and Smatch++ add softer or more standardized graph comparison variants [14, 15]. These metrics are important for parser evaluation, but they presuppose graph representations on both sides. LEMON-FACTOR instead evaluates factor evidence from graph-text pairs and only uses reconstructed graphs in the reverse experiment.

2.4 Text-to-KG extraction and MINE

Recent text-to-KG work has renewed interest in evaluating whether extracted graphs retain useful information from plain text. KGGen introduces a language-model-based text-to-KG generator and releases MINE, Measure of Information in Nodes and Edges, as a benchmark for testing whether an extractor produces useful KGs from text [16]. Our MINE-STYLE baseline is inspired by this

Table 1. Positioning of LEMON-FACTOR relative to related evaluation families.

Family	Representative methods	Granularity	Missing diagnostic dimension	Direction
Surface/reference	BLEU, METEOR, chrF++, TER, BERTScore, BLEURT	sentence-level holistic	source-predicate roles	KG→Text
Source-aware	PARENT, QuestEval, Align-Score	Data- n-gram, question, sentence	explicit within-predicate factors	KG→Text
Triple-level	FactSpotter	binary triple	which part of the predicate failed	KG→Text
Atomic fact	FactScore, VERISCORE	text-derived atomic claims	source predicate decomposition	Text→KB
Node/edge	KGGen/MINE	node and edge retention	weighted predicate factors	Text→KG
LEMON-FACTOR	this work	weighted factors inside predicates	provisional resource, not human gold	bidirectional

node/edge information split. It is not a reproduction of the KGGen benchmark, which targets KG extractors on plain text; rather, it adapts the idea to WebNLG graph-text pairs so that Text→KG recoverability can be compared with Reverse LEMON-FACTOR.

2.5 Behavioral testing and metric calibration

Evaluation metrics should be tested not only by correlation with human judgments, but also by whether they respond correctly to controlled changes. CheckList introduced behavioral tests for NLP models, including invariance tests, where label-preserving perturbations should not change predictions, and directional expectation tests, where perturbations should change predictions in a prescribed direction [17]. Perturbation CheckLists for NLG metrics apply the same idea to automatic metric evaluation [18]. He et al. use synthetic stress tests to expose blind spots of model-based NLG metrics [19]. Our calibration follows this methodology: relation deletion is a directional test, while entity aliasing and predicate paraphrase templates are invariance-style checks whose quality is explicitly marked.

3 Method

3.1 Problem setting

Let a graph-text pair be (G, T) , where G is a set of directed edges $e = (s, p, o)$ and T is a verbalization. The forward task asks whether the text covers the information in the graph. The reverse task asks whether a graph reconstructed from the text, denoted $\hat{G}(T)$, preserves the information in the original graph. LEMON-FACTOR uses the same predicate-level semantic representation in both directions.

3.2 Predicate factorization

Each predicate p is mapped to a weighted decomposition:

$$D(p) = \{(f_i, r_i, w_i)\}_{i=1}^n,$$

where f_i is a semantic factor, r_i is its role, and $w_i > 0$ is a salience weight. Roles include *predicate meaning*, *subject domain*, *object domain*, *value domain*, and modifier or background roles. The schema uses broad factors such as *person*, *place*, *organization*, *time*, *quantity*, and relation factors such as *biographical relation*, *location relation*, *part-whole relation*, *political relation*, and *creative relation*. This representation is intentionally coarser than a full ontology but richer than a single predicate label.

For example, *birthPlace* receives a person-like subject factor, a biographical predicate factor, and a place-like object factor. In the biomedical inventories, *chemical_inhibits_gene_or_protein* receives chemical and gene/protein role factors, an inhibition factor, negative polarity, directionality, and evidence-form factors. A text that mentions both entities but omits the relation cue should therefore receive partial credit, not full coverage. This is the key distinction from whole-triple binary scoring.

3.3 Forward LEMON: KG→Text coverage

For an edge e , text T , and component (f_i, r_i, w_i) , an evidence function $E(f_i, r_i, e, T) \in [0, 1]$ estimates whether the component is expressed. In the deterministic pilot, entity and domain factors are supported by normalized label or token evidence, while relation factors are supported by predicate cues and factor cues. The edge-level forward score is:

$$S_F(e, T) = \frac{\sum_i w_i E(f_i, r_i, e, T)}{\sum_i w_i}.$$

Example-level and corpus-level scores are averages over edges with available decompositions. The implementation stores covered and missing factors for each edge, so the scalar score is accompanied by a diagnostic trace.

3.4 Reverse LEMON: Text→KG recoverability

The reverse direction first constructs a deterministic reconstructed graph $\hat{G}(T)$ from the text using entity-label evidence and predicate lexical cues. Reverse LEMON-FACTOR then compares each gold edge with its best recovered counterpart. Subject and object recovery contribute through node/domain evidence; predicate recovery contributes through overlap between the gold predicate’s factors and the recovered edge evidence. The reverse score therefore asks whether the source predicate’s semantic factors survive the text-to-graph reconstruction.

The current reconstructed graph is a lexical pseudo-extractor, not a full KG extraction model. This choice makes the pilot reproducible and keeps the focus on metric behavior rather than extractor quality. Future work can replace $\hat{G}(T)$ with KGen, GraphRAG-style extraction, OpenIE, or domain-specific extractors.

Table 2. Bidirectional evaluation pipeline.

Stage	Input and operation	Output
Predicate factors	source predicate p is decomposed into weighted factors	role-aware semantic representation
Forward LEMON	compare factors of G with evidence in T	KG→Text coverage and missing-factor trace
Reverse LEMON	reconstruct $\hat{G}(T)$, compare gold and recovered factors	Text→KG recoverability trace
MINE-style	retrieve nodes/edges for gold facts from $\hat{G}(T)$	node, edge, composite, and fact-recoverability scores
Calibration	inject controlled text/graph perturbations	monotonicity and invariance diagnostics

3.5 MINE-style node/edge baseline

To compare factor-level recoverability with node/edge information retention, we implement a MINE-STYLE baseline inspired by KGGen’s MINE. Each gold triple is treated as a fact. The reconstructed graph is searched for matching or related nodes and edges. The deterministic score separates node information, edge information, and a composite partial-credit node/edge score. A stabilized LLM-judged variant receives a gold fact and a compact retrieved subgraph and returns a binary fact-recoverability judgment. We report these quantities separately because a partial-credit node/edge composite and a binary fact judgment are not the same metric.

3.6 Controlled perturbation calibration

A metric that measures semantic alignment should degrade under meaning-destroying perturbations and remain comparatively stable under meaning-preserving perturbations. We therefore create deterministic perturbations of both text and reconstructed graphs. Text-side destructive perturbations include relation phrase deletion, object swaps, and predicate swaps. Graph-side destructive perturbations include node drops, edge drops, and predicate swaps. Preserving perturbations include alias-like rewrites and template paraphrases, which are flagged as clean or weak-template depending on their reliability. All perturbed files and operation-level manifests are saved, enabling audit of each injected change.

4 Data and Experimental Resources

4.1 WebNLG pilot

The experiments use a stratified English WebNLG pilot. The train split contains 200 examples and the development split contains 100 examples. Both splits cover 16 WebNLG categories. The development set contains 308 graph edges. The pilot is intentionally small: it validates the end-to-end scoring pipeline, exposes interpretable failure modes, and allows manual inspection of perturbation manifests before scaling to larger WebNLG subsets or domain-specific graphs.

Table 3. Pilot WebNLG statistics.

Split	Examples	Edges	Avg. edges	Predicates	Categories
Train	200	568	2.84	162	16
Development	100	308	3.08	119	16

Table 4. Fixed predicate-factor inventories after LEM-03. Coverage is computed against the current evaluation split used by the next experiments.

Dataset	Factors	Predicates	Data predicates	Coverage
webnlg	24	121	119	1.000
drugprot	25	13	12	1.000
bc5cdr	25	1	1	1.000

4.2 Predicate decomposition resources

LEM-03 fixes the predicate-factor inventories used by the final experiments. The resource is stored as a deterministic JSON decomposition set: a controlled factor vocabulary plus one weighted decomposition per predicate. WebNLG uses the current fixed development inventory. DrugProt adds a biomedical inventory for chemical-gene/protein relations, including activation, inhibition, agonist, antagonist, regulator, substrate, product, and part-of predicates. BC5CDR uses a compact document-level inventory for the chemical-induced disease relation. The biomedical inventories make polarity, directionality, participant roles, and evidence form explicit, which is needed for the perturbation and ablation studies.

A synthetic adjudicator was applied to 29 predicates. It modified 19 decisions, accepted 10 seed decisions, and produced a mean confidence of 0.830. LLM candidate coverage was 0.621 because 18 of the 29 adjudicated predicates had LLM candidates. The resulting resource is useful for developing the framework, but it remains synthetic and provisional.

5 Experiments

The experiments are organized by the four research questions from the introduction rather than by development stage.

5.1 RQ1: Forward factor-level semantic coverage

We first score WebNLG development texts using the synthetic adjudicated predicate decompositions and seed lexical cues. We then run an error-driven expansion: missing predicates are identified, rule-based or fallback decompositions are added, lexical cues are expanded, and coverage is recomputed. This measures how much of the initial low score comes from missing resources rather than from the text itself.

Table 5. LLM decomposition candidate quality against the seed reference. This is a sanity check, not human validation.

Model	Items	Factor P	Factor R	Factor F1	Role Acc.
Claude 3.5 Haiku	19	0.754	0.807	0.775	0.596
Llama 3.1 70B Instruct	19	0.798	0.798	0.795	0.614

Table 6. Synthetic adjudication diagnostics.

Metric	Value
Items	29
LLM candidate coverage	0.621
Predicates with LLM candidates	18
Predicates without LLM candidates	11
Mean confidence	0.830
Modified decisions	19
Accepted seed decisions	10

5.2 RQ2: Bidirectional factor recoverability

To test whether the factor representation is bidirectional, we reconstruct a graph from each WebNLG text with a deterministic lexical pseudo-extractor and compute Reverse LEMON-FACTOR. This score is compared with expanded Forward LEMON-FACTOR. A small forward–reverse gap would suggest that the same factor schema can support both $\text{KG} \rightarrow \text{Text}$ coverage and $\text{Text} \rightarrow \text{KG}$ recoverability.

5.3 RQ3: MINE-style node/edge information retention

We compare Reverse LEMON-FACTOR with a MINE-STYLE baseline inspired by KGGen’s Measure of Information in Nodes and Edges. The deterministic variant reports node information, edge information, and a composite node/edge score. The LLM-judged variant evaluates binary fact recoverability on a fixed 50-fact subset using compact JSON-constrained judgments. We report parse success, deterministic agreement, and confidence to avoid treating LLM judgments as unqualified gold.

5.4 RQ4: Controlled perturbation calibration

We inject deterministic noise into text and reconstructed graphs. Destructive perturbations should reduce relevant scores; preserving perturbations should produce smaller changes. All perturbed examples, reconstructed graphs, and operation-level manifests are stored under a separate perturbation directory. To avoid misleading averages, text-side perturbations are used primarily to calibrate Forward LEMON-FACTOR, while graph-side perturbations are used primarily to calibrate Reverse LEMON-FACTOR and MINE-style recoverability.

LEM-04 adds a shared perturbation interface for WebNLG, DrugProt, and BC5CDR. The operators keep the source graph fixed and change only the text,

Table 7. Controlled perturbation operators introduced in LEM-04. The operators keep the source graph fixed and alter the text so that factor-level sensitivity can be tested in LEM-05.

Variant	Target factor groups	Expected effect
Node deletion	Entity presence; participant roles	Removes one endpoint mention and should reduce node and role evidence.
Edge deletion	Predicate meaning; evidence form	Removes a relation cue while preserving the graph, so entity overlap may stay high.
Argument swap	Directionality; participant roles	Swaps subject and object mentions and should damage role assignment.
Polarity flip	Polarity; predicate meaning	Replaces positive, negative, or causal cues with conflicting cues.
Relation blur	Predicate specificity; evidence form	Replaces a specific relation cue with a broad cue such as <i>affects</i> or <i>is associated with</i> .

which lets us test whether a metric reacts to the intended semantic damage rather than to a change in the reference graph. Table 7 summarizes the five operators used by the final calibration and ablation runs. Polarity flips are most informative for DrugProt and BC5CDR, where relation labels explicitly encode activation, inhibition, regulation, or causality.

5.5 RQ5: Cross-domain baseline scoring

Finally, we score the shared perturbation files with deterministic baselines. The goal is not to reproduce external systems such as FactSpotter or KGGen/MINE, but to provide local comparison points with the same inputs. We report entity recall, label match, a binary triple-style match, a MINE-STYLE node/edge harmonic score, a label-only LEMON variant, and the full deterministic LEMON-FACTOR proxy based on the fixed factor inventories and perturbation metadata. This step provides the scoring layer for later ablation and LLM-adjudication experiments.

6 Results

6.1 RQ1: Forward coverage and resource expansion

The initial factor dictionary covered only part of the WebNLG development predicates. As a result, the first coverage run scored 140 of 308 edges and produced a missing decomposition rate of 0.5455. Error-driven expansion reduced the missing rate to zero and increased LEMON-FACTOR coverage from 0.3683 to 0.7915 (Table 8). This increase is not presented as proof that all texts are faithful; it shows that factor-level evaluation is sensitive to the completeness of predicate decompositions and lexical evidence.

The comparison with lexical controls (Table 9) clarifies the signal. Exact entity-label overlap is already high and unchanged by expansion. Graph-text token cosine and Jaccard are also unchanged. LEMON-FACTOR, in contrast,

Table 8. Forward LEMON-FACTOR before and after error-driven decomposition and cue expansion.

Metric	Initial	Expanded	Delta
Scored edges	140	308	+168
Missing decomposition rate	0.5455	0.0000	-0.5455
Predicate cue coverage	0.3149	0.6526	+0.3377
LEMON-FACTOR coverage	0.3683	0.7915	+0.4232
Low-coverage edges	191	41	-150

Table 9. Baseline comparison on the WebNLG development pilot.

Method	Initial	Expanded	Delta
Exact entity-label overlap	0.9215	0.9215	0.0000
Predicate lexical cue coverage	0.3149	0.6526	+0.3377
Graph-text token cosine	0.7357	0.7357	0.0000
Graph-text token Jaccard	0.6233	0.6233	0.0000
LEMON-FACTOR coverage	0.3683	0.7915	+0.4232

changes because it uses predicate decompositions and role-aware evidence. This supports the interpretation that LEMON-FACTOR measures a different diagnostic layer than surface overlap.

6.2 RQ2: Bidirectionality

The reverse experiment recovered 390 of 408 nodes and 196 of 308 edges. Reverse LEMON-FACTOR reached 0.7789, close to the expanded Forward LEMON-FACTOR score of 0.7915 (Table 10). The small gap suggests that the predicate-factor representation can be used in both directions: from graph to text and from text back to graph. The result should be interpreted as a pilot observation because the reconstructed graph is produced by a lexical pseudo-extractor.

6.3 RQ3: Relation information is harder than node information

The full deterministic MINE-STYLE composite score is 0.8028, but this aggregate hides a strong node/edge gap. On the locked 50-fact subset (Table 11), node information is 0.9600, while edge information and deterministic fact recoverability are both 0.5400. The LLM-judged fact recoverability score is 0.5200, with 0.9800 agreement with the deterministic fact decision and 100% parse success. These numbers indicate that WebNLG texts tend to preserve entity mentions more reliably than relation recoverability. They also show why the partial-credit composite score and the binary LLM fact score must not be treated as interchangeable.

6.4 RQ4: Perturbation calibration

Table 12 reports targeted calibration summaries after separating text-side and graph-side perturbations. Destructive text perturbations produce a monotonic

Table 10. Bidirectional comparison on the WebNLG development pilot.

Method	Direction	Factors	Retrieval	Score	Diagnostic level
Forward LEMON-FACTOR	KG→Text	yes	no	0.7915	factor/edge/example
Reverse LEMON-FACTOR	Text→KG	yes	no	0.7789	factor/edge/example
MINE-style composite node/edge	Text→KG	no	yes	0.8028	node/edge/fact
MINE-style LLM fact recoverability	Text→KG	no	yes	0.5200	fact/subgraph
Graph-text token cosine	KG↔Text	no	no	0.7357	corpus
Graph-text token Jaccard	KG↔Text	no	no	0.6233	corpus

Table 11. MINE-style score semantics on the fixed 50-fact subset.

Metric	Value
Facts	50
Deterministic node information	0.9600
Deterministic edge information	0.5400
Deterministic composite node/edge score	0.7500
Deterministic fact recoverability	0.5400
LLM fact recoverability	0.5200
Judge agreement with deterministic	0.9800
Parse success rate	1.0000
Mean judge confidence	0.5417

Forward LEMON-FACTOR decrease, while destructive graph perturbations produce monotonic decreases in Reverse LEMON-FACTOR and MINE-style recoverability. This supports the expected directional behavior of the metrics.

The clearest sanity check is relation phrase deletion (Table 13). Exact entity-label coverage changes by only -0.0020, while Forward LEMON-FACTOR drops by -0.2106. This is the central validity signal of the pilot: entity preservation alone is not relation preservation, and the factor-level metric reacts to relation-level semantic omission.

6.5 RQ5: Cross-domain deterministic scoring

Table 14 reports compact deterministic scores on the shared perturbation files introduced in LEM-04. Full metric outputs are written to JSON and Markdown reports. Entity recall remains relatively high for relation blur and polarity changes, especially in BC5CDR, because the same biomedical entities are still present. In contrast, LEMON-FACTOR full decreases when the edited text targets participant roles, directionality, polarity, or predicate evidence. This is the expected behavior of a factor-level diagnostic: it assigns damage to the affected semantic components rather than only to surface entity loss.

The WebNLG polarity-flip row should be read with care. Many WebNLG predicates in the current pilot do not encode a polarity factor, so the deterministic factor proxy does not penalize that edit strongly. This is useful rather than problematic: it shows that the factor inventory controls which semantic dimensions can be tested. The biomedical rows are more informative for polarity and direction because DrugProt and BC5CDR contain relation labels where those dimensions are part of the task definition.

Table 12. Targeted perturbation calibration. Slopes are average score changes with increasing destructive noise.

Metric	Target	Destr. slope	Spearman	Viol.	Pres. delta
Forward LEMON-FACTOR	text	-0.3241	-1.0000	0	0.0527
Exact label coverage	text	-0.1508	-0.9162	1	0.0585
Predicate cue coverage	text	-0.4525	-0.6667	0	0.0438
Reverse LEMON-FACTOR	graph	-0.5546	-1.0000	0	0.0000
MINE composite node/edge	graph	-0.5161	-1.0000	0	0.0000
MINE fact recoverability	graph	-0.7204	-1.0000	0	0.0000

Table 13. Relation deletion sanity check.

Metric	Baseline	Final	Delta
Exact label coverage	0.9215	0.9194	-0.0020
Forward LEMON-FACTOR	0.7915	0.5808	-0.2106
Predicate cue coverage	0.6526	0.2890	-0.3636
MINE fact recoverability	0.6364	0.2825	-0.3539

Table 14. Compact deterministic scoring summary over controlled perturbations.

Dataset	Perturbation	Entity	Triple	LEMON-full
WebNLG	Node deletion	0.5142	0.0857	0.6603
WebNLG	Relation cue deletion	0.7287	0.1643	0.7031
WebNLG	Argument swap	0.7212	0.2977	0.7080
WebNLG	Polarity flip	0.7237	0.3010	1.0000
WebNLG	Relation blur	0.7237	0.1643	0.7031
DrugProt	Node deletion	0.5281	0.1459	0.7000
DrugProt	Relation cue deletion	0.5867	0.1604	0.7543
DrugProt	Argument swap	0.5869	0.1869	0.6019
DrugProt	Polarity flip	0.5872	0.1774	0.5285
DrugProt	Relation blur	0.5867	0.1604	0.7543
BC5CDR	Node deletion	0.8321	0.5288	0.6000
BC5CDR	Relation cue deletion	0.9659	0.5412	0.6924
BC5CDR	Argument swap	0.9696	0.7280	0.5500
BC5CDR	Polarity flip	0.9667	0.7222	0.5512
BC5CDR	Relation blur	0.9659	0.7208	0.6924

7 Error Analysis

The experiments reveal two recurring error classes. The first is resource incompleteness: predicates without decompositions or with weak lexical cues cannot be scored reliably. The error-driven expansion step was designed to expose and reduce this problem. After expansion, all 308 development edges had decompositions, but fallback decompositions remain less informative than expert-designed ones.

The second class is relation recoverability. Node information is high because WebNLG verbalizations often repeat entity labels, but edge information is substantially lower. This pattern appears in the reverse experiment, in the MINE-style node/edge gap, and in the relation-deletion perturbation. It suggests that the difficult part of graph-text alignment is not usually naming entities, but preserving and detecting the relation that links them.

The factor-level trace is useful here. A low LEMON-FACTOR can be traced to a predicate, a factor role, a missing cue, or a weak subject/object evidence signal. This is the main practical advantage of the framework: it turns a scalar score into a debugging signal for datasets, generators, and extraction pipelines.

8 Limitations

The main limitation is the absence of human expert validation. The current factor decompositions are produced by rules, LLM candidates, and synthetic LLM adjudication. They should be read as a bootstrap resource, not as expert gold. Work on LLM-as-a-judge evaluation reports systematic biases, so the factor schema and adjudicated decompositions require future expert annotation and inter-annotator agreement studies [23–25].

A second limitation is lexical evidence. The deterministic scorer can find labels and known cues, but it can miss implicit relations, long-range entailment, and paraphrases outside the cue dictionary. Future versions should add embedding, entailment, or QA-based evidence while preserving the factor-level audit trail.

A third limitation is the reverse pipeline. The reconstructed graph is produced by a lexical pseudo-extractor, not by a full text-to-KG system. Likewise, our MINE-style baseline adapts KGGen’s node/edge information idea to WebNLG graph–text pairs; it is not a reproduction of the original MINE benchmark. The deterministic MINE-style composite score, deterministic fact recoverability, and LLM fact recoverability measure related but distinct quantities.

A fourth limitation is perturbation validity. The controlled perturbations are deterministic and auditable, but some meaning-preserving perturbations are template-like. They support initial construct-validity checks, especially relation deletion, but they do not prove robustness to the full range of natural paraphrase. Larger studies should use curated paraphrase sets and human validation of perturbation labels.

Finally, the present experiments are WebNLG-only and use a small stratified pilot. This is appropriate for framework development, but it does not justify broad claims about domain transfer. Biomedical and other domain-specific experiments require domain-specific factor inventories, explicit entity and relation mapping, and a separate reliability analysis. In the repository plan, DrugProt and BC5CDR are treated as transfer diagnostics, not as evidence for a new state-of-the-art biomedical relation extraction system.

9 Conclusion

We introduced LEMON-FACTOR, a bidirectional factor-level diagnostic framework for graph–text pairs. The method decomposes RDF predicates into weighted semantic factors and evaluates both KG→Text coverage and Text→KG recoverability. In a WebNLG pilot, expanded Forward LEMON-FACTOR reached 0.7915

Table 15. Terminological distinctions used to avoid overclaiming.

Term	Meaning	Status in this pilot
Human gold	expert factor judgments under annotation guidelines	not collected
Seed/rule reference	decompositions from rules and schema constraints	used for bootstrapping
Synthetic LLM adjudication	LLM-generated or LLM-adjudicated decompositions	provisional, not expert validation
MINE-style adaptation	WebNLG adaptation of node/edge information retention	not full KGGen/MINE reproduction
Deterministic style score	MINE-lexical node/edge recoverability without LLM judgment	transparent baseline

and Reverse LEMON-FACTOR reached 0.7789, which supports the use of one predicate-factor representation in both directions.

The MINE-style comparison showed that node information is easier to recover than edge information. Controlled perturbation calibration showed that relation deletion strongly reduces factor-aware scores while leaving exact entity-label coverage nearly unchanged. These findings position LEMON-FACTOR as a diagnostic layer between triple-level factuality and atomic-fact evaluation. Future work will replace synthetic adjudication with expert annotation, add stronger paraphrase evidence, and test the same factor-level design on domain-specific graph-text data, including biomedical relations.

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